An extended certificate of small size is decided by the matchings it forces

Gilad Chase, Neta Dafni, Yuval Filmus, Nathan Lindzey

arxiv:2210.00245v2; Creative Commons Attribution 4.0 International

Editorial note. The counted statement is the author's labelled environment at source0.tex lines 694-696 and its proof environment opens at line 697; the pairing is by adjacency and the retained proof range is the whole environment.

Editorial note. Preamble: the author's own package lines and theorem-environment family (source lines 5-24); the layout packages of the source are not retained because the retained material does not use them.

Editorial note. Retained-licence form: the licence evidence of this source carries no acceptance record, and none may be written on anyone's behalf, so the card points at a task-directory evidence copy in the team-wide four-key reuse form (name, version_url, prior_license_path, basis) whose basis states that the source's staged record is reused as a material basis only. No status key and no acceptance record are written.

Editorial note. Metadata provenance: source_version is the value recorded in the source's licence evidence (2210.00245v2); the author list is transcribed from the authors_index key of the source record of the pinned copy, and the title from its title key.

Editorial note. The source carries no label-number dump and no counter values are imposed, so the printed numbering is the wrapper's default; the identity of the counted item is its author label.

Editorial note. r002 note: r001 failed with an undefined control sequence because the counted statement uses the author's macro for the family of perfect matchings, and the author's macro block (source lines 26-29) had not been retained; r002 adds exactly that block as a further preamble span and changes nothing else.

Editorial scope. One original result of the pinned author manuscript, counted once: the consequence of an extended certificate of small size (author label lem:extended-certificates).

Allowed standard prerequisites (editorial).

- Extended certificate. A certificate together with one additional triple of indices, recording that a matching satisfies the pairs and the triple in a prescribed way.. It is the object whose consequence is asserted.

- Satisfying a certificate. A perfect matching satisfies a certificate when it contains all its pairs, and it satisfies an extended one when in addition the triple is arranged as prescribed.. It is the relation between the two hypotheses of the statement.
- Pair representation. A matching satisfying a small certificate consists of that certificate together with several further pairs, which leaves enough room for two flips.. It is the bound that guarantees the two alternative matchings exist.

Lemma 0.1. *Fix $n \geq 1$, work over $\mathcal{M}_{2n}$, and let $c \leq n-2$. Let $C_1 = (C_1', \{i,j,k\})$ be an extended certificate of size c, and let C_2 be an extended certificate of size c. If any perfect matching satisfying C_1 and any perfect matching satisfying C_2 are 2-intersecting, then the same holds when C_1 is replaced by C_1'.*

Proof. Suppose that m_1 is a perfect matching satisfying C_1' but not C_1. Without loss of generality, $m_1(i) = j$. We need to show that if m_2 is a perfect matching satisfying C_2 then m_1 and m_2 are 2-intersecting.

Since m_1 satisfies C_1, its pair representation consists of C_1 together with $n-(c-1) \geq 3$ more pairs, one of which is $\{i,j\}$. Among the other pairs, at least one $\{a,b\}$ does not involve k. Consider the following two perfect matchings:

$$m_1' = m_1 \setminus \big\{\{i,j\},\{a,b\}\big\} \cup \big\{\{i,a\},\{j,b\}\big\},$$
$$m_1'' = m_1 \setminus \big\{\{i,j\},\{a,b\}\big\} \cup \big\{\{i,b\},\{j,a\}\big\}.$$

Both of these perfect matchings avoid the edges $\{i,j\},\{i,k\},\{j,k\}$. If m_2 contains $\{i,a\}$ then it cannot contain $\{i,b\}$ or $\{j,a\}$, and similarly for $\{j,b\},\{i,b\},\{j,a\}$. Therefore m_2 cannot intersect both $\big\{\{i,a\},\{j,b\}\big\}$ and $\big\{\{i,b\},\{j,a\}\big\}$. Suppose that m_2 does not intersect $\big\{\{i,a\},\{j,b\}\big\}$. Then $m_1' \cap m_2 \subseteq m_1 \cap m_2$. On the other hand, m_1' satisfies C_1, and so $|m_1' \cap m_2| \geq 2$. We conclude that $|m_1 \cap m_2| \geq 2$, as needed. □